Lecture 11: Support Vector Machines

Binary classification

- Given training data (x_i, y_i) for i = 1, 2, …, N where each y_i is -1 or +1
- You want to learn a classifier f(x) such that:

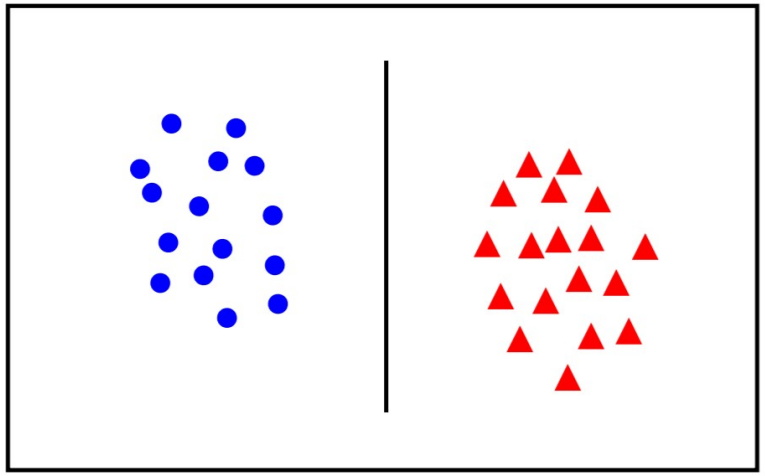
$$f(\mathbf{x}_i) \begin{cases} \geq 0 & y_i = +1 \\ < 0 & y_i = -1 \end{cases}$$

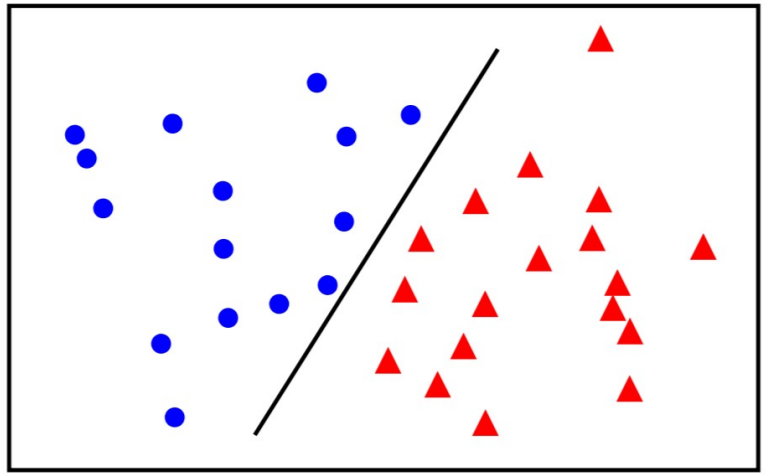
Math is largely subjective, but this has the advantage of:

$y_i * f(x_i) > 0$ for a correct classification

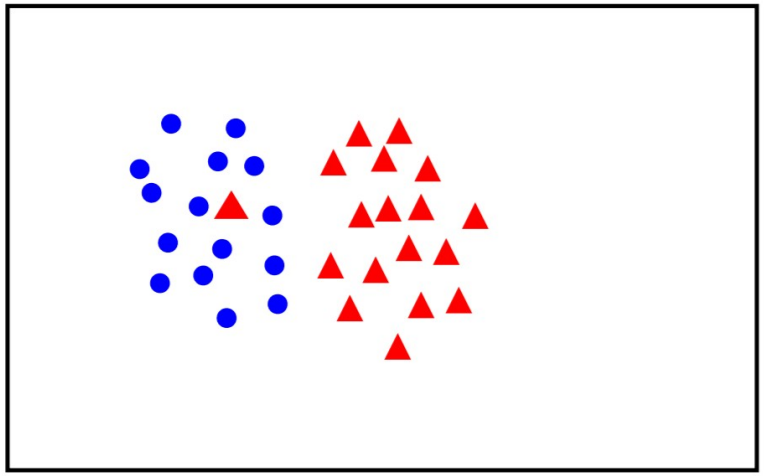
Linear separability

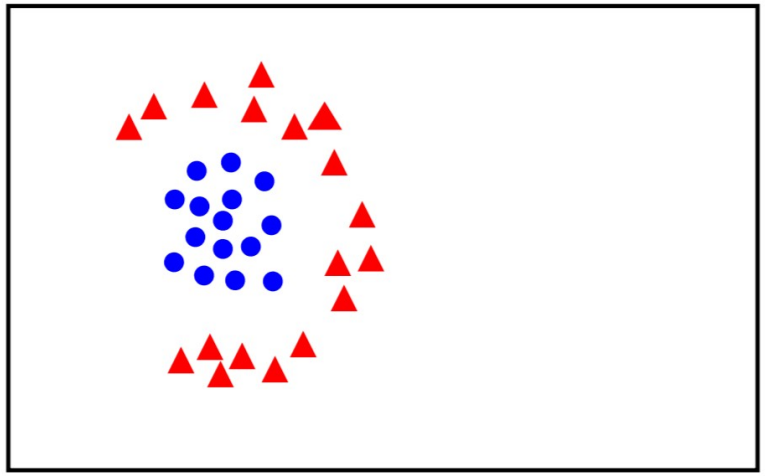
linearly separable





not linearly separable

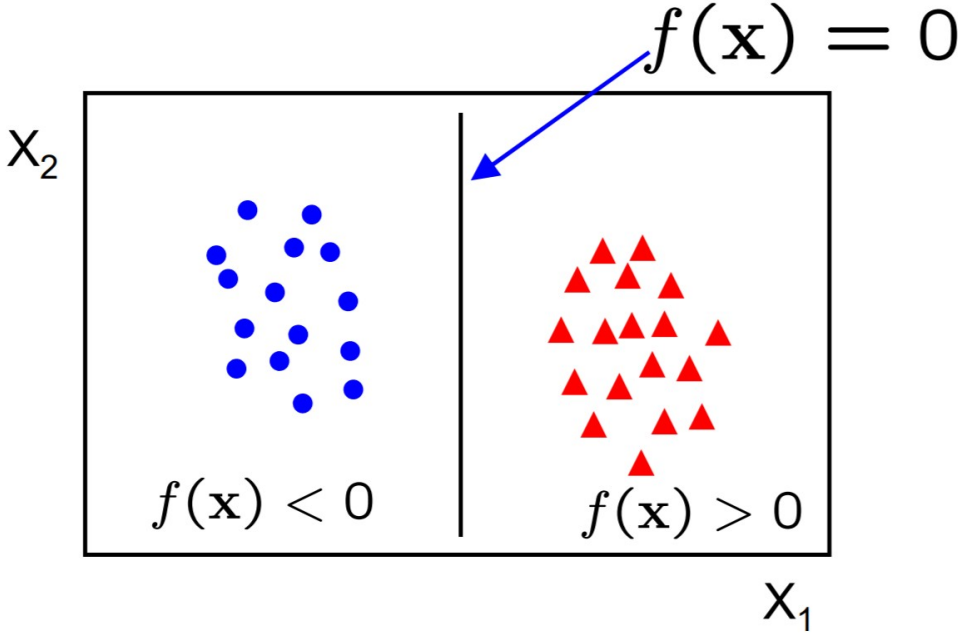




Linear classifiers

- A linear classifier has the form $f(x) = w^T x + b$

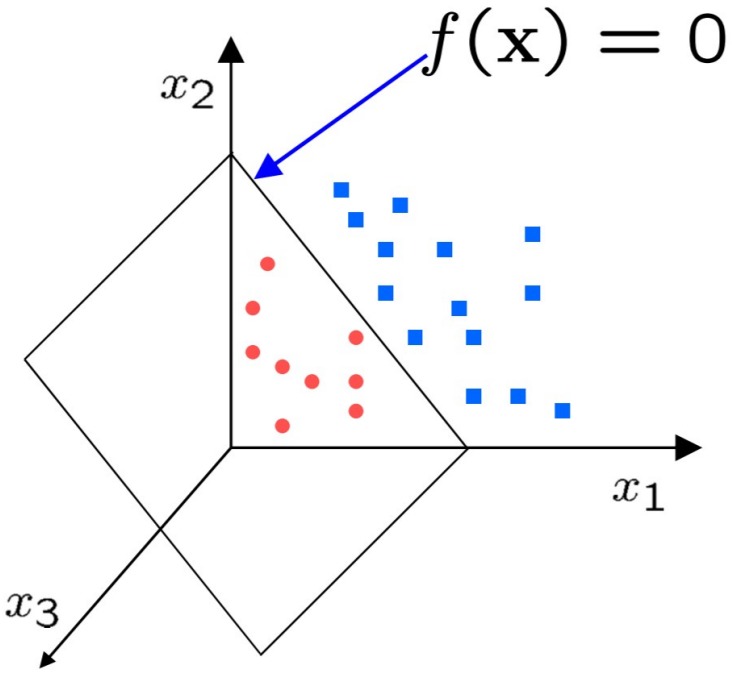
- In 2D, discriminant is a line
- w is the normal to the line, b is the bias
- w is known as the weight vector



Linear classifiers

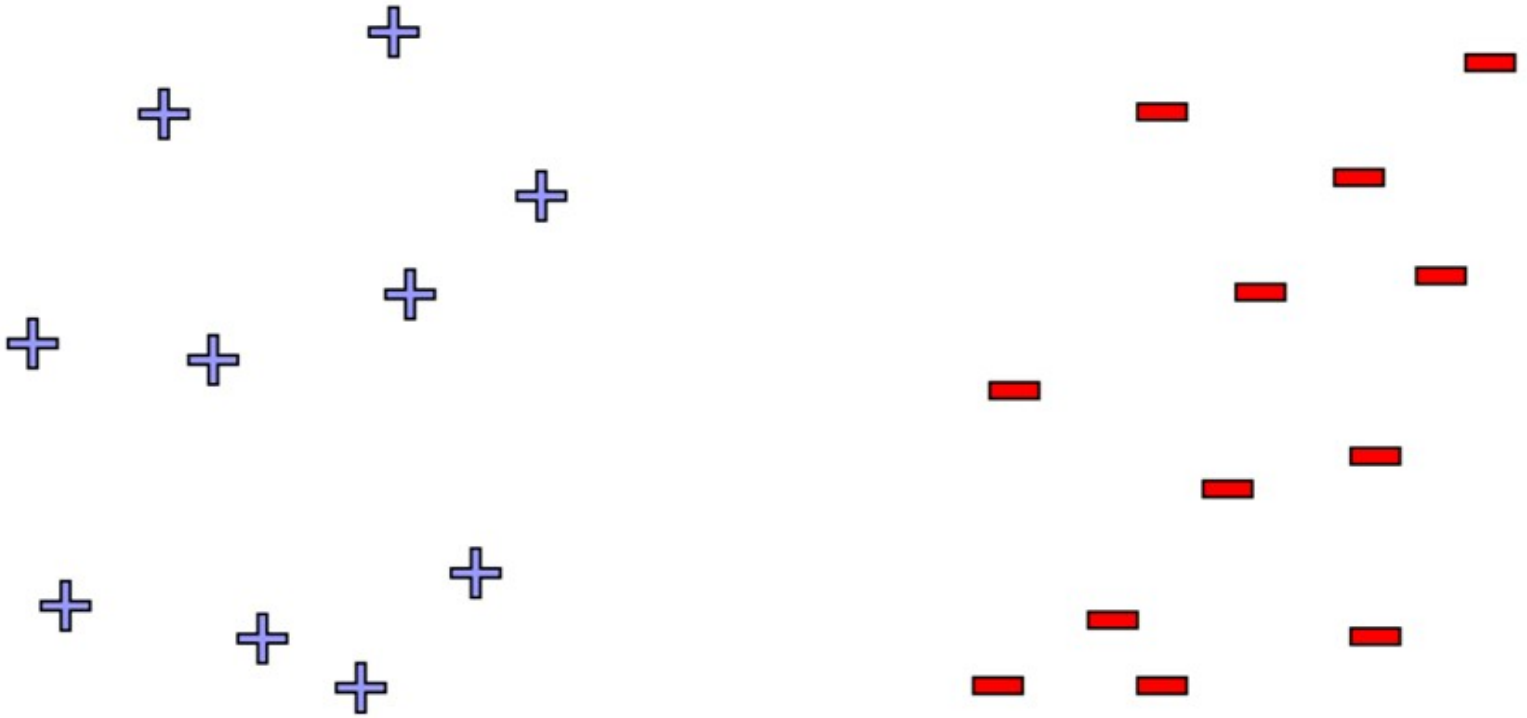
- A linear classifier has the form $f(x) = w^T x + b$

- In 3D, discriminant is a plane
- For a linear classifier, the training data is used to learn w and then discarded
- Only w is needed for classifying new data

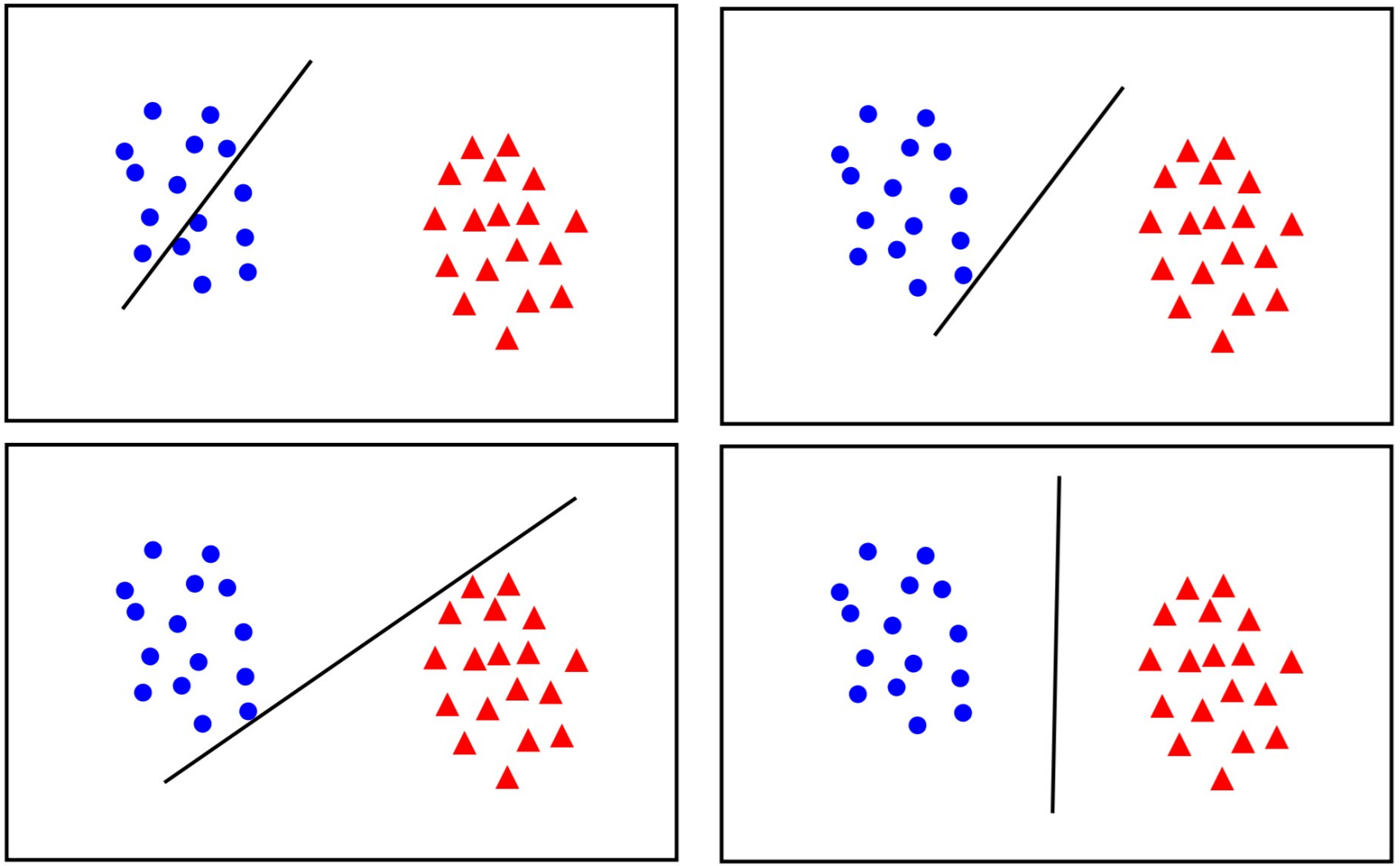


Reminder: perception algorithm

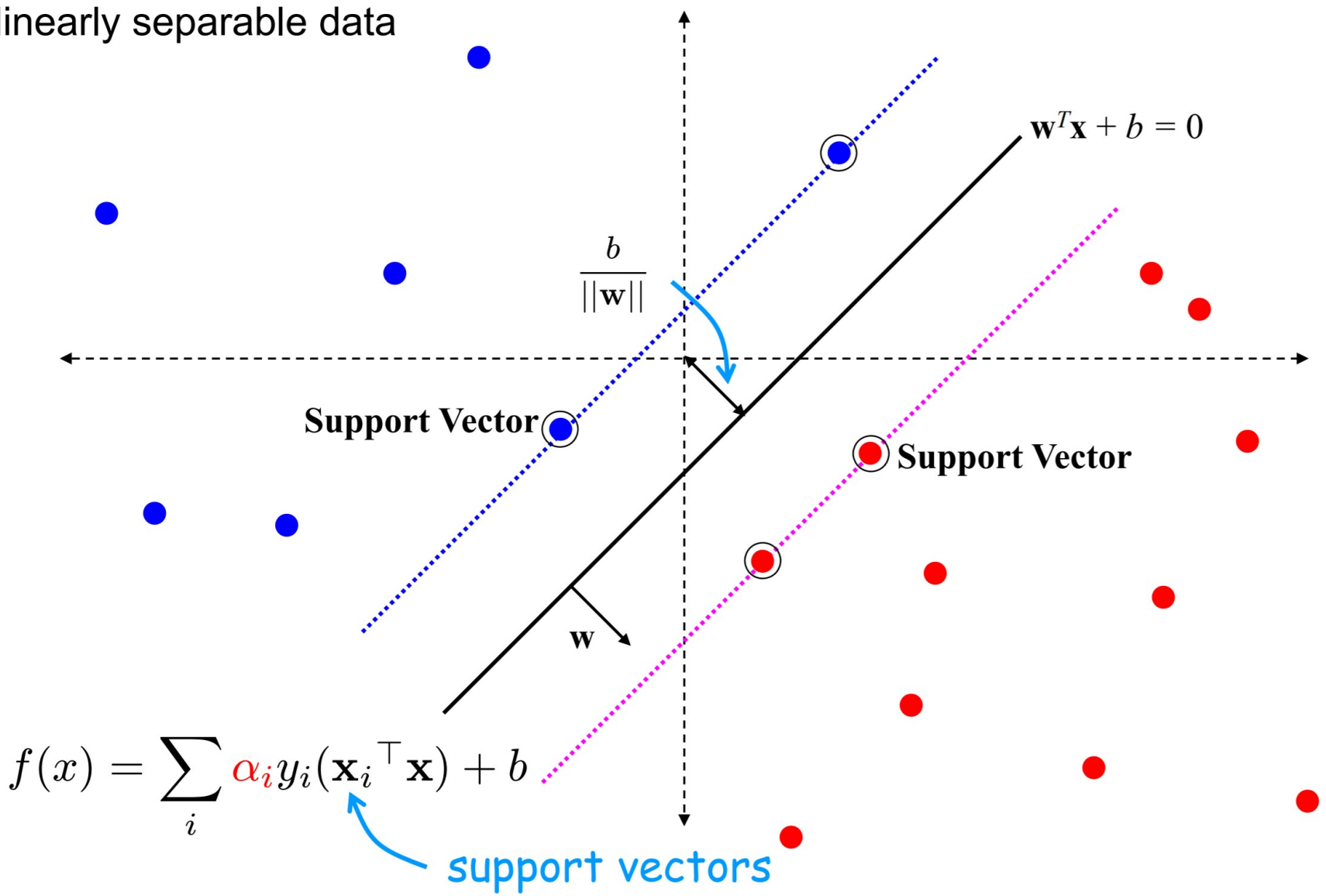
- Perception is guaranteed to converge as long as the data is linearly separable



Perception algorithm – where to draw the line?



Support **V**ector **M**achine (SVM)

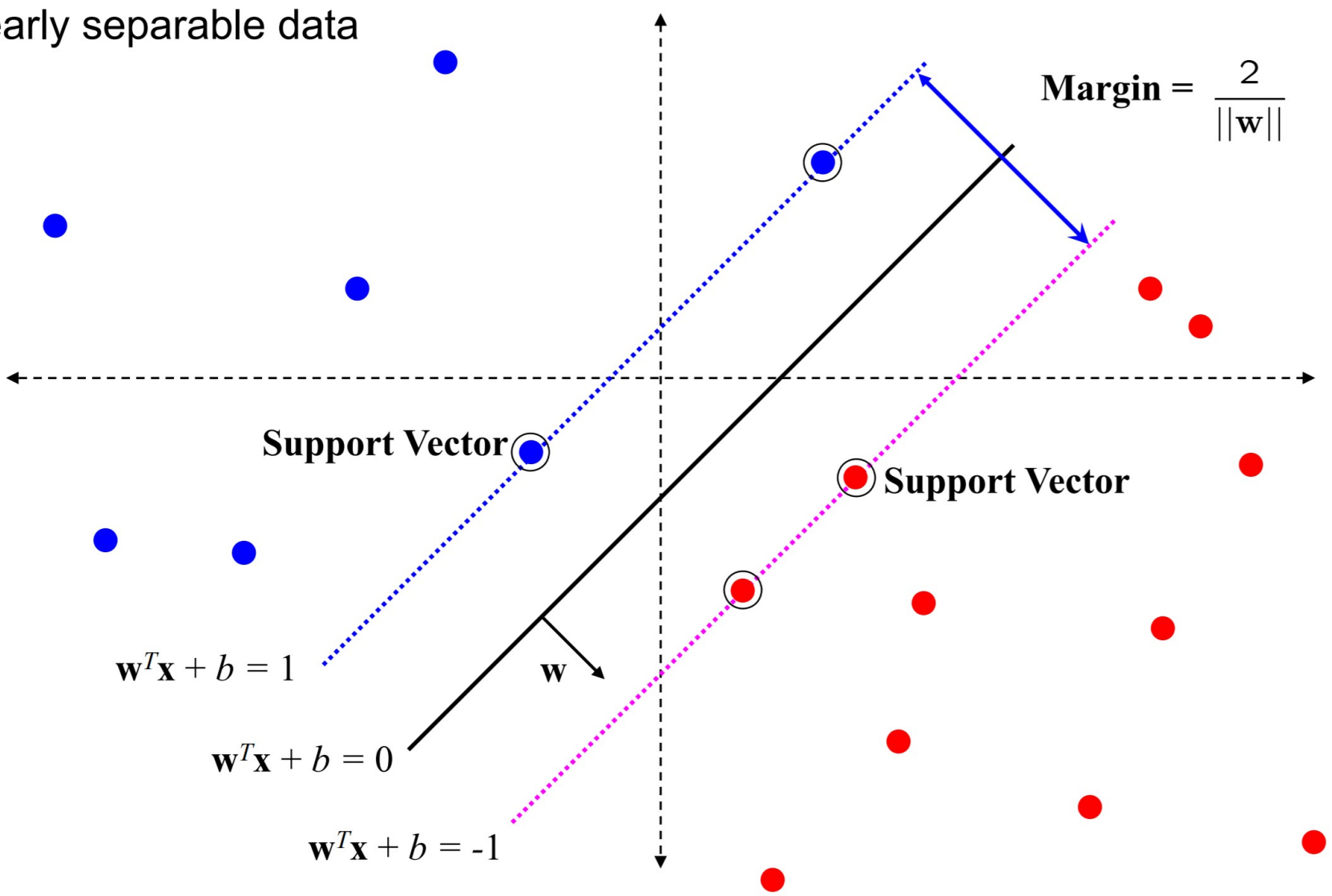


SVM

- Since $w^T x + b = 0$ and $c(w^T x + b) = 0$ define the same plane, we have the freedom to choose the normalization of w
- Choose normalization such that $w^T x + b = +1$ and $w^T x + b = -1$ for the positive and negative support vectors respectively
- Then the margin is given by:

$$\frac{\mathbf{w}}{||\mathbf{w}||} \cdot \left(\mathbf{x}_+ - \mathbf{x}_-\right) = \frac{\mathbf{w}^\top \left(\mathbf{x}_+ - \mathbf{x}_-\right)}{||\mathbf{w}||} = \frac{2}{||\mathbf{w}||}$$

SVM


linearly separable data
Margin = 2 / ||w||
Support Vector
Support Vector
w
$\mathbf{w}^T\mathbf{x} + b = 1$
$\mathbf{w}^T\mathbf{x} + b = 0$
$\mathbf{w}^T\mathbf{x} + b = -1$

SVM -- optimization

- Learning the SVM can be formulated as an optimization:

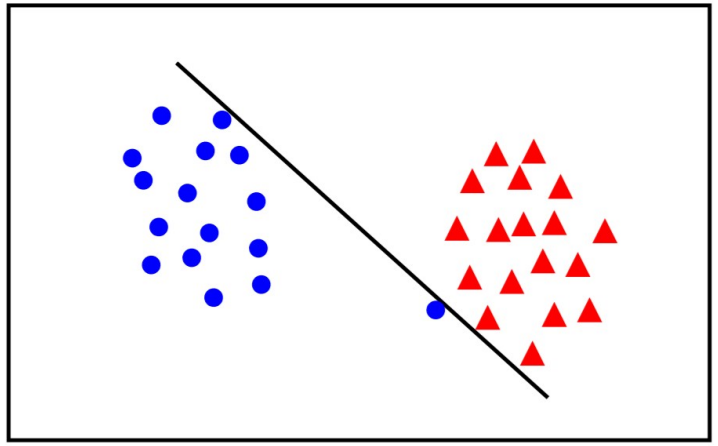
$$\max_{\mathbf{w}} \frac{2}{||\mathbf{w}||} \text{ subject to } \mathbf{w}^\top \mathbf{x}_i + b \begin{array}{ll} \geq 1 & \text{if } y_i = +1 \\ \leq -1 & \text{if } y_i = -1 \end{array} \text{ for } i = 1 \ldots N$$

- Or equivalently:

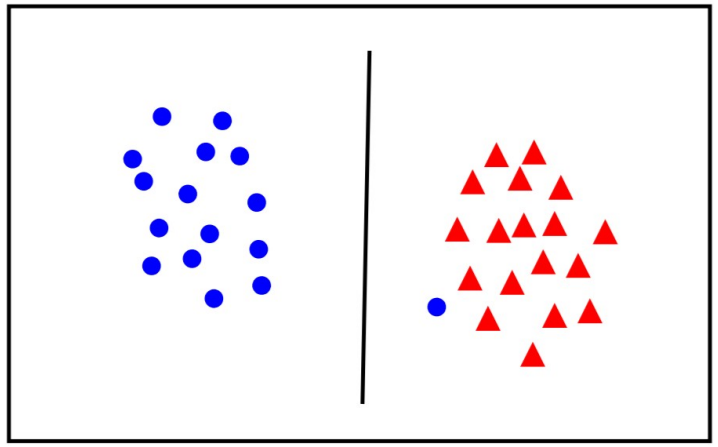
$$\min_{\mathbf{w}} ||\mathbf{w}||^2 \text{ subject to } y_i \left(\mathbf{w}^\top \mathbf{x}_i + b \right) \geq 1 \text{ for } i = 1 \ldots N$$

- This is a quadratic optimization problem subject to linear constraints and there is a unique minimum

Selecting the best w



• the points can be linearly separated but there is a very narrow margin



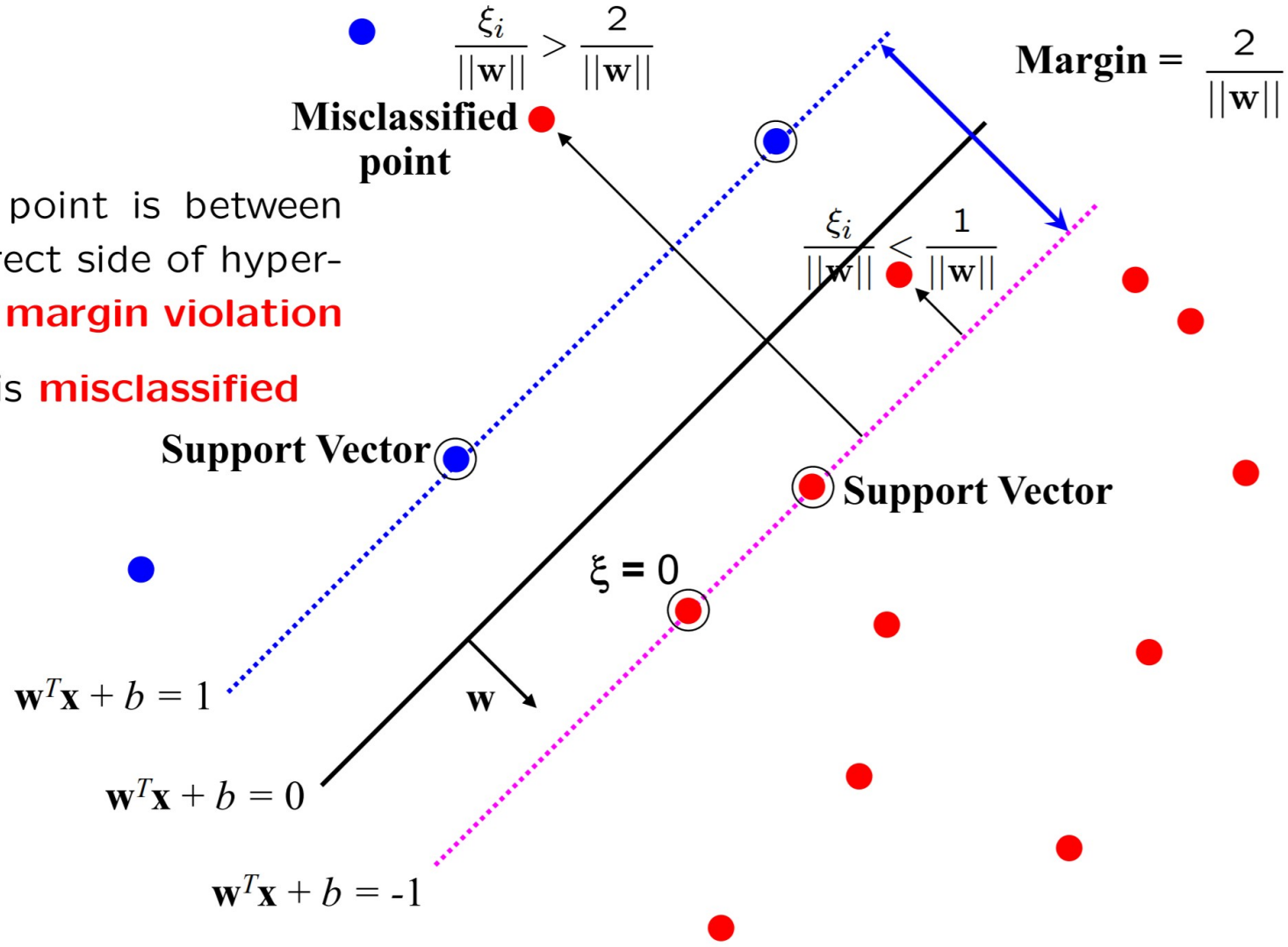
• but possibly the large margin solution is better, even though one constraint is violated

In general there is a trade off between the margin and the number of mistakes on the training data

“Slack” variables

$$\xi_i \geq 0$$

- for $0 < \xi \leq 1$ point is between margin and correct side of hyper-plane. This is a **margin violation**
- for $\xi > 1$ point is **misclassified**



“Soft” margin solution

The optimization problem becomes

$$\min_{\mathbf{w}\in\mathbb{R}^d,\xi_i\in\mathbb{R}^+} ||\mathbf{w}||^2+C\sum_i^N \xi_i$$

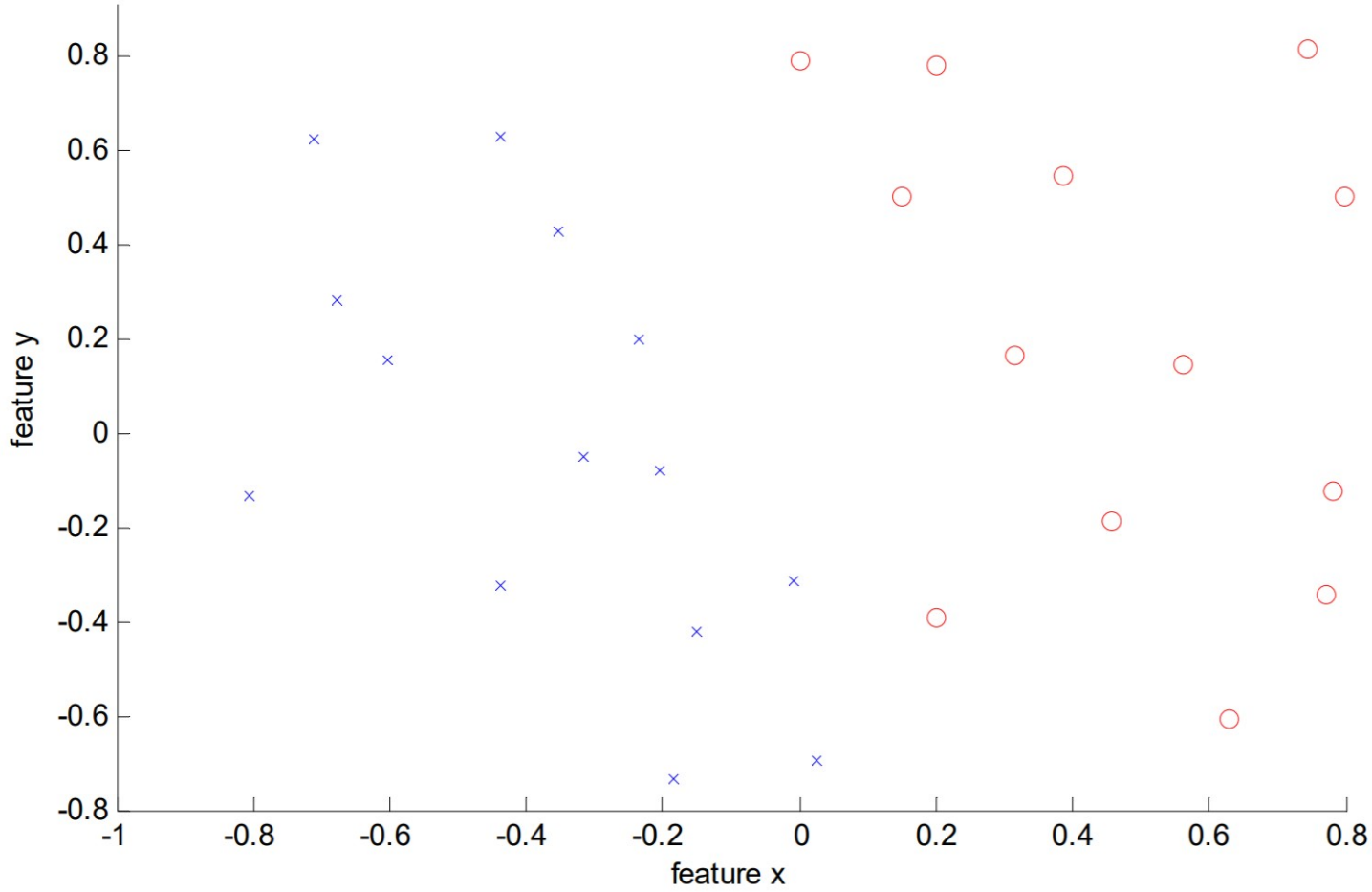
subject to

$$y_i\left(\mathbf{w}^\top\mathbf{x}_i+b\right)\geq 1-\xi_i \text{ for } i=1\ldots N$$

- Every constraint can be satisfied if ξ_i is sufficiently large
- C is a regularization parameter:
  - small C allows constraints to be easily ignored $\rightarrow$ large margin
  - large C makes constraints hard to ignore $\rightarrow$ narrow margin
  - $C=\infty$ enforces all constraints: hard margin
- This is still a quadratic optimization problem and there is a unique minimum. Note, there is only one parameter, C.

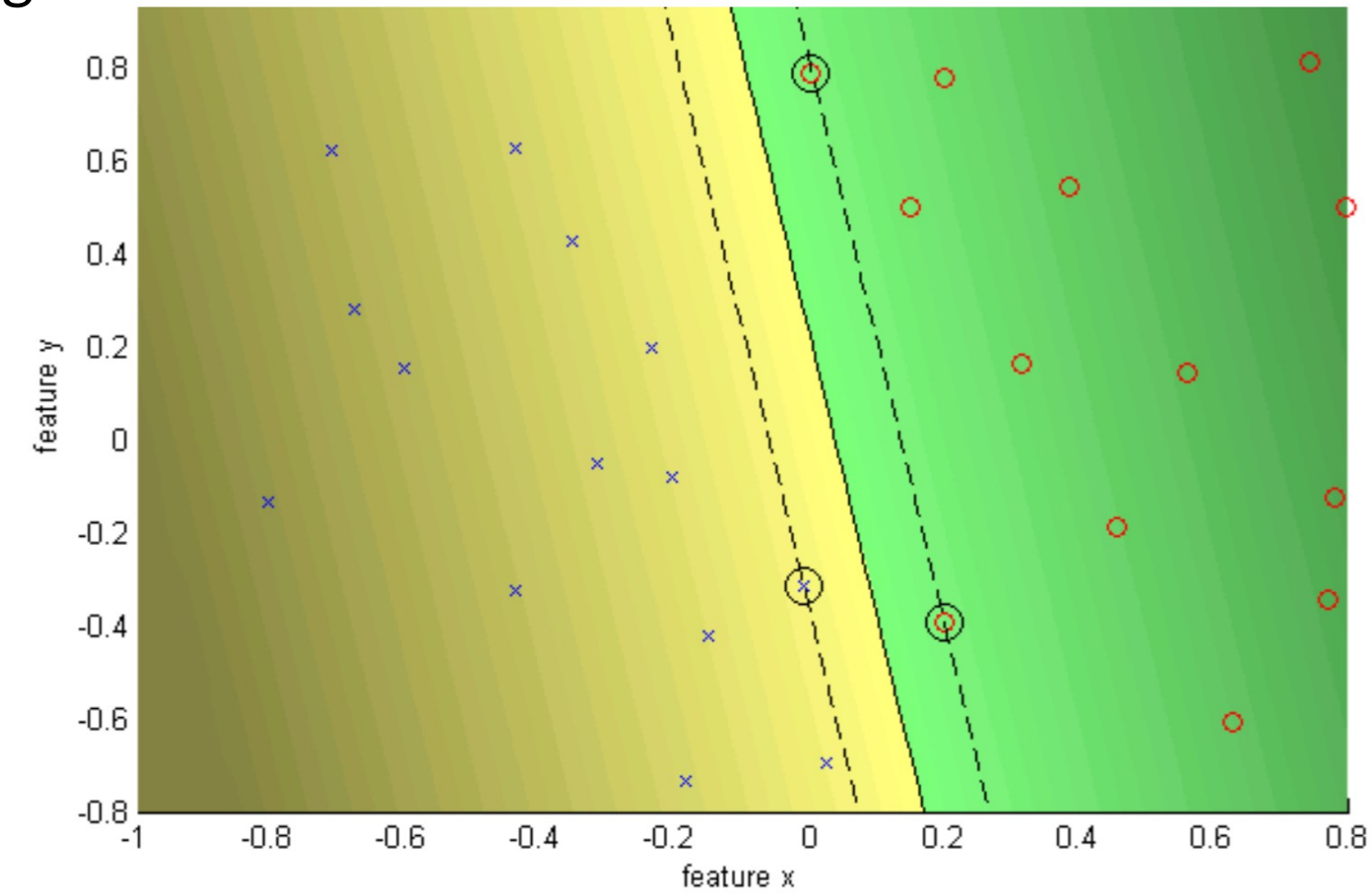
SVM

- Data is linearly separable, but only in a narrow margin



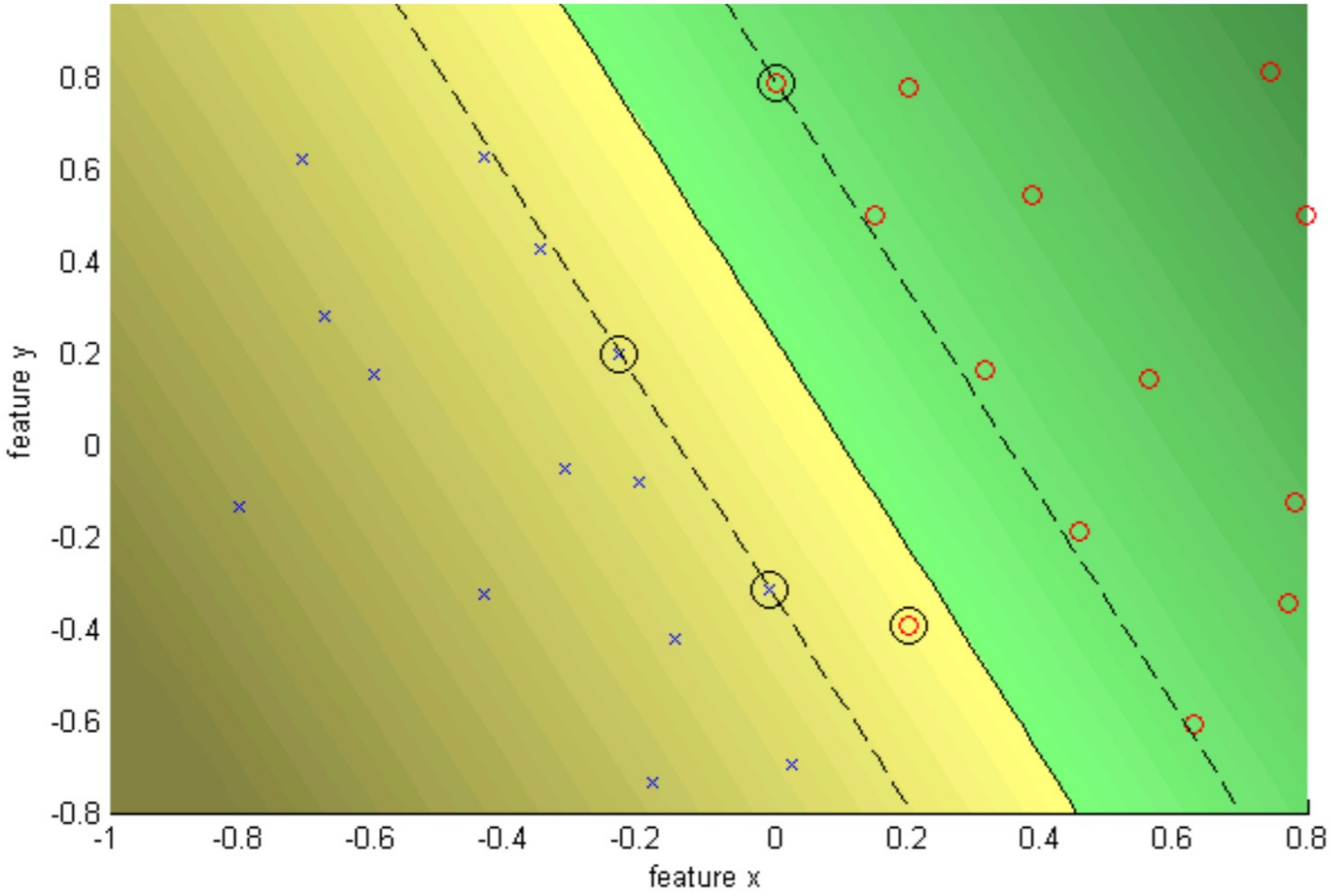
SVM – C = infinity

- Hard margin



SVM – C = 10

- Soft margin



Soft vs hard margin

- Also sometimes called “soft constraint” and “hard constraint”
- Refers to whether or not all of the training data is “correctly” classified by the classifier
  - If all “dots” sort as they should, it’s a hard constraint
  - If there are “dots” in the inappropriate region, it’s a soft constraint
  - The presence of “slack” makes a soft margin

Application: pedestrian detection in computer vision

- Objective: detect (localize) standing humans in an image (face detection with a sliding window classifier)
  - Reduces object detection to binary classification
  - Does an image window contain a person or not?

Training data and features

Positive data:



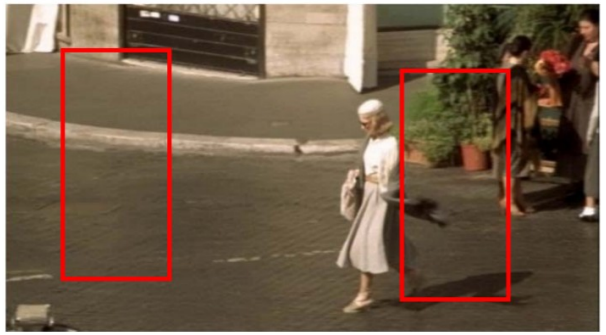






Negative data:





SVM “big picture”

- We want to maximize the margin over the training data – that is the line should be maximally far away from the most number of data points

- This means that you essentially want to minimize w
  - This optimum w is often termed w^*

SVM “big picture”

- Variables:
  - Cost parameter, C
    - Too large, may have overfitting
    - Too small, may have underfitting
  - Epsilon, ε, determines how far away from the margin the support vectors are
    - You want enough support vectors to get good results, not so many that it takes long…